

The 4th & Goal Puzzle

STAT 4800 - Final Project

Maximilian Van Zandt, Andrew Tran, Zain Bangash, and JJ Sutkus

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1. Introduction

As statisticians, sports fans, and all professionals in the National Football League (NFL) agree, understanding how coaching decisions affect team outcomes is paramount in putting professional athletes in a position to win for their franchise. This paper explores just one type of decision that coaches make: late-game decisions to “go for it” or kick a field goal on 4th & Goal.

We ask the following central question in our study:

In a 4th & Goal situation, in what scenarios should we go for it versus kick a field goal and what factors are necessary for making this decision?

This question interests us because it involves a high-impact decision during a key moment of the game and has various elements that affect the probability of success. NFL coaches have recently experimented more with 4th-down plays, particularly the Washington Commanders’ coach, Dan Quinn, who is notorious for his 4th-down attempts. Through his strategy, the Washington Commanders clinched the NFC Championship for the first time since the 1990s. The reason why coaches don’t always go for 4th-down, however, is that not all 4th-down attempts are made equal. Some 4th-downs are extremely risky, while others are no-brainers. This is compounded by the fact that 4th & Goal is one of the most pivotal points in a football game that can flip momentum, impact win probability, and determine whether a coach is strategic or reckless. When thinking about decision-making models for these situations, the probability of success can be attributed to many things: the score differential, time remaining, distance from the endzone, offensive and defensive formations, and more. The combination of these factors makes it worth exploring as the decision made on 4th & Goal can certainly sway the outcome of the game, affecting the reputation of the coach as a decision-maker.

This paper is a statistical simulation study that combines multiple modeling approaches into one framework. Below, we discuss this framework.

2. Methodological Framework

In this section, we discuss two aspects of our study: 1) We discuss the original simulation that forms the foundation with which we can begin to ask our research question; 2) We discuss the modifications we’ve made to this simulation and argue why these modifications are appropriate for answering our central research question.

2.1 Original Simulation

The original simulation this study is based off of models such as the field goal success probability model, the fourth down decision model, the expected points model, the decision criterion, the yards gained mixture

model, and the expected value of a possession. These models were developed in class, but they are not necessarily catered towards our specific application. They model separate decisions within a game and can be built upon. Nevertheless, we provide the formulas of these models below to demonstrate the baseline framework with which we worked off of.

Field Goal Success Probability Model

For field goal attempts, we implemented a logistic regression model.

$$P(\text{FG Success}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \cdot \text{distance})}}$$

Where:

- β_0 and β_1 are the coefficients from the logistic regression model
- $\text{distance} = 120 - \text{fp}$ (field position)

This estimates the probability of successfully making a field goal from a given field position.

Fourth Down Decision Model

We implemented a multinomial regression model for 4th down decisions. The probability of choosing each option (go for it, punt, or field goal) can be expressed as:

$$P(\text{decision} = j | \text{fp}, \text{ytg}) = \frac{e^{\beta_{j0} + \beta_{j1} \cdot \text{fp} + \beta_{j2} \cdot \text{ytg}}}{\sum_{k=1}^3 e^{\beta_{k0} + \beta_{k1} \cdot \text{fp} + \beta_{k2} \cdot \text{ytg}}}$$

Where:

- $j \in \{\text{go_for_it}, \text{punt}, \text{field_goal}\}$
- $\beta_{j0}, \beta_{j1}, \beta_{j2}$ are the coefficients for decision j from the multinomial model

Expected Points Model

For 4th & Goal situations, the expected points when going for it can be formulated as:

$$EP(\text{go_for_it}) = P(\text{success}) \cdot 7 + (1 - P(\text{success})) \cdot EP(\text{opp_possession})$$

Where:

- $P(\text{success})$ is the probability of converting the 4th down into a touchdown
- $EP(\text{opp_possession})$ is the expected points when opponent takes possession at the goal line

Decision Criterion

The optimal decision rule for 4th & Goal situations:

$$\text{Decision} = \begin{cases} \text{Go for it,} & \text{if } EP(\text{go_for_it}) > EP(\text{field_goal}) \\ \text{Field goal,} & \text{otherwise} \end{cases}$$

Where:

$$EP(\text{field_goal}) = P(\text{FG Success}) \cdot 3 + (1 - P(\text{FG Success})) \cdot EP(\text{opp_possession})$$

Yards Gained Mixture Model

We use a mixture of normal distributions to model yards gained. For each play type (run or pass) and field zone:

$$P(\text{yards gained} = y) = \sum_{i=1}^k \lambda_i \cdot \phi(y; \mu_i, \sigma_i^2)$$

Where:

- λ_i is the weight of component i in the mixture
- $\phi(y; \mu_i, \sigma_i^2)$ is the normal probability density with mean μ_i and variance σ_i^2
- k is the number of components in the mixture

Expected Value of a Possession

The expected value of a possession starting at down d , yards to go y , and field position f can be recursively defined as:

$$V(d, y, f) = \sum_g P(g|d, y, f) \cdot V(T(d, y, f, g))$$

Where:

- g is the yards gained on the play
- $P(g|d, y, f)$ is the probability of gaining g yards given the state (d, y, f)
- $T(d, y, f, g)$ is the transition function that gives the next state after gaining g yards

2.2. Overview of Modifications and Justification

After discussing the key components of our original simulation, we move onto how we can tailor these models to help us answer our central research question. Most 4th & Goal situations are at critical points in the game (usually end-of-game scenarios), which made us come to the conclusion that win probability would be more important than expected points. The reason is that expected points may be negligible at certain points if the team does not actually increase their chance of winning. This is especially apparent in late-game scenarios. For example, while there may be higher expected points if a team goes for a field goal over a risky touchdown attempt, a high score differential combined with little time may mean that a field goal is simply not enough points. More on this kind of scenario in the next section. For now, consider that this made us tailor our model more to win probability as opposed to expected points.

We also incorporated some key factors we thought would have the greatest impact on win probability as it relates to 4th & Goal decision-making. Specifically, time remaining in the game is a key factor in coaching decisions on whether or not to be aggressive, depending on how likely a team is to get the ball back. Point differential is another determinant in making the decision to go for it or not, as whether you are up or within a field goal, touchdown, or it is a two-possession game in either direction will certainly sway the decision-making regarding the fourth down. Yards to go also impacts this decision, as it drastically sways the likelihood of success on 4th & Goal depending on the distance to the endzone. We determined that these factors were vital for making these decisions on 4th & Goal, and adding them to our model would result in a more successful win probability model.

2.3 Modifications

Given these considerations, we made the following modifications to the decision criterion to prioritize win probability over expected points, and consider that within certain time frames. We also adjusted the win probability model to account for end-game scenarios.

Win Probability Change Model

For each potential decision outcome, the win probability change can be expressed as:

$$\Delta WP = WP_{new} - WP_{current}$$

Where:

- WP_{new} is the win probability after the play
- $WP_{current}$ is the win probability before the play

This calculation allows us to consider how a particular decision improves win probability, rather than simply quantifying the probability itself. Specifically, by isolating the value added by a particular decision, this helps us control the game state before a particular decision is made, offering more nuanced guidance.

Base Win Probability Model

The base win probability as a function of score differential and time remaining:

$$WP(score_diff, time_remaining) = \min(1, \max(0, 0.5 + score_impact))$$

Where $score_impact$ is calculated as:

$$score_impact = \left(1 - e^{-|score_diff|/7}\right) \cdot \text{sign}(score_diff) \cdot 0.5$$

This model calculates how win probability changes based on score differential and remaining time, starting with a neutral 0.5 probability and then adjusting based on the score impact. We can see that the formula uses diminishing returns approach where each additional point of score differential has less impact than the previous point. We calibrated the denominator of the exponent by a factor of 7 which reflects win probabilities in NFL games. Finally, the sign function ensures the positive score differentials increase win probability.

For end-game scenarios (time remaining < 5 minutes), the score impact is adjusted:

$$score_impact_{adjusted} = \begin{cases} \min(0.5, score_impact \cdot 1.5) & \text{if } score_diff > 0 \\ \max(-0.5, score_impact \cdot 1.5) & \text{if } score_diff < 0 \end{cases}$$

This adjustment reflects our desire to increase the importance of score differential late in games. With less than 5 minutes remaining, the chance a team will have multiple possessions decreases significantly. This means that the score differential at that point in the game becomes more indicative of what the final outcome of the game may be. More specifically, we added a 1.5 multiplier to amplify the effect of the score differential. Also note that the min/max constraints make sure that the probability remains within reasonable bounds. Contextually, this reflects game situations where teams that are leading by many points see their win probability increase dramatically in the last 5 minutes of a game while the team that is trailing by many points sees the chances of making a comeback dwindle.

Decision Model for 4th and Goal

The modified decision criterion that incorporates game context:

$$Decision = \begin{cases} \text{"Go for it"} & \text{if } (score_diff \in [-8, -4] \text{ and } time_remaining \leq 5) \\ \text{"Kick field goal"} & \text{if } (score_diff \in [-2, -1] \text{ and } time_remaining \leq 2 \text{ and } P(FG) > 0.8) \\ \text{"Go for it"} & \text{if } \Delta WP_{go} > \Delta WP_{fg} \\ \text{"Kick field goal"} & \text{otherwise} \end{cases}$$

Where:

- ΔWP_{go} is the expected change in win probability from going for it
- ΔWP_{fg} is the expected change in win probability from attempting a field goal
- $P(FG)$ is the probability of making the field goal

Note that the intervals reflect our previous discussion of situations where the score differential is in a range such that a field goal (3 points) would not be sufficient to close the score differential. This is what the first condition handles. We included the second condition because, within 2 minutes, if the score differential is enough to where a field goal would make up the difference, a field goal should be recommended as it is generally less risky (especially if the $P(FG \text{ success}) > 0.8$) than relying on a 4th down conversion to a touchdown. Moreover, we chose to target the last 2 minutes in this condition because the chances of a team having two separate possessions within 2 minutes are incredibly low. Thus, the 4th & Goal decision made within 2 minutes must assume that this is the team's last chance to secure points.

3. Results

Our simulation results demonstrate how different variables affect optimal 4th & Goal decision-making.

Figure 1 shows different recommendations for different 4th & Goal scenarios, given the expected value of scoring a touchdown versus the expected value of scoring a field goal. This illustrates the kind of raw output a coach may receive in the moment. The scenarios show simulations for recommendations on a tied game and a game where one is down by a touchdown, spread across the final 5 minutes and 10 minutes of a game. Being able to identify scenarios that maximize expected points can be imperative for securing points, however, that is not always a recipe for success. Win probability is ultimately the thing that far outweighs any other metric and is by far the most significant factor in coaching decisions in 4th & Goal situations. Interestingly, the simulation results show that, in tie games, yards to go matters significantly, with consistent Kick field goal' recommendations for longer attempts (5+ yards). Conversely, when down by a touchdown (7 points) with less than 5 minutes remaining, yards to go matters much less. Still though, yards to go is an important factor influencing decisions in close games and thus, should still be considered.

Figure 2 shows the percentage of which we should go for it in different game scenarios, segmented by time remaining and the score differential. The green line represents the average of "Go for it" recommendations in the final 5 minutes—the most important part of a game, regardless of score differential. We can see though this that it is not until around 30 minutes left in the game, or the start of the second half, that we begin to see noticeable and meaningful change in percentage in which going for it on fourth down is recommended. This highlights how the state of the game in the first half is not as impactful as it is down the stretch and in late-game scenarios, which is the main time frame of the game we tried to hone in on in our analysis. Specifically, the dramatic drop in 'Go for it' recommendations between the 5-minute and 15-minute marks reveals a critical inflection point in decision-making strategy. This confirms our suspicions that the final minutes of a game fundamentally changes win probability and subsequently, decision-making, especially in situations where teams are down by 7 points (blue line).

In Figure 3, the left panel provides us with an order of magnitude on when we should go for it instead of kicking a field goal on the basis of time remaining (from 0 seconds to 60 seconds) and score differential. This plot provides a blueprint for coaching decisions on fourth down situations based on how much time is left in the game and the current point differential for the team in the fourth down scenario.

The right panel of Figure 3 shows the fourth-down recommendations by score differential. This highlights some interesting strategic ideas surrounding when a team should be aggressive versus remaining conservative. For instance, being down one or two points results in the lowest recommendation for going for it due to the high probability of converting a field goal which will result in a team taking the lead. This is a lower probability of a tie game or being up by one or three which is an important thing to note. This is because converting on fourth down, and hence scoring a touchdown, will place immense pressure on the other team, creating a scenario where they will almost certainly need a touchdown, especially late in the game.

Overall, Figure 3 suggests that score differential is the primary driver of decisions. Specifically, we can see through the heatmap that the deepest blue regions appear at high negative score differentials regardless of time remaining. This pattern directly corresponds with the right panel's bar chart, where we see 'Go for it' recommendations peaking at approximately 70% for teams down by 10-14 points. These two patterns confirm that point deficit outweighs other factors in these critical situations.

Figure 4 displays the win probability differential between kicking a field goal and scoring a touchdown based on the time remaining and the number of yards to the end zone, segmented across different score differentials. Knowing how a win probability differs between kicking a field goal and going for it at key times and point differentials in the game can be crucial for team success and maximizing opportunity. Being aware of the smallest differences in win probability could be the difference in a team winning or losing games. Specifically, the win probability curves at a threshold of around 2-4 yards to goal which is more pronounced in late-game scenarios (green lines) than in mid-game situations (orange lines). This shows how time pressure intensifies the impact of each decision on win probability.

Figure 5 presents a recommendation of whether or not to go for it or kick a field goal plotted against the number of yards we have to go in our 4th & Goal as well as the score differential. It is also segmented

across the number of minutes remaining from 1 minute to 5 minutes. Having these recommendations in the final 4 minutes provides us a concrete cutoff of which decisions to make and when, allowing us the greatest opportunity to win. Overall, these decision charts show relatively consistent patterns across all time frames, with green boxes (recommending teams ‘go for it’) concentrated at areas of high score differential and low yards to go. Notably, there appears to be a decision boundary at around 3 yards to go.

Scenario: 4th and goal from the 1 yard line, score differential: 0 , time remaining: 10 minutes
 Expected points if going for it: 4.35
 TD success probability: 62.8 %
 Expected points if kicking field goal: 2.47
 FG success probability: 88.9 %
 Recommendation: Go for it

Scenario: 4th and goal from the 2 yard line, score differential: 0 , time remaining: 10 minutes
 Expected points if going for it: 2.77
 TD success probability: 40.6 %
 Expected points if kicking field goal: 2.55
 FG success probability: 88.4 %
 Recommendation: Go for it

Scenario: 4th and goal from the 5 yard line, score differential: 0 , time remaining: 10 minutes
 Expected points if going for it: 1.72
 TD success probability: 25.8 %
 Expected points if kicking field goal: 2.46
 FG success probability: 86.9 %
 Recommendation: Kick field goal

Scenario: 4th and goal from the 10 yard line, score differential: 0 , time remaining: 10 minutes
 Expected points if going for it: 1.14
 TD success probability: 17.6 %
 Expected points if kicking field goal: 2.37
 FG success probability: 84.1 %
 Recommendation: Kick field goal

Scenario: 4th and goal from the 3 yard line, score differential: -7 , time remaining: 5 minutes
 Expected points if going for it: 2.2
 TD success probability: 32.7 %
 Expected points if kicking field goal: 2.48
 FG success probability: 87.9 %
 Recommendation: Kick field goal

Scenario: 4th and goal from the 3 yard line, score differential: 7 , time remaining: 5 minutes
 Expected points if going for it: 2.08
 TD success probability: 31 %
 Expected points if kicking field goal: 2.56
 FG success probability: 87.9 %
 Recommendation: Kick field goal

Figure 1: 4th & Goal simulations showing success probabilities and expected points for touchdown vs. field goal options.

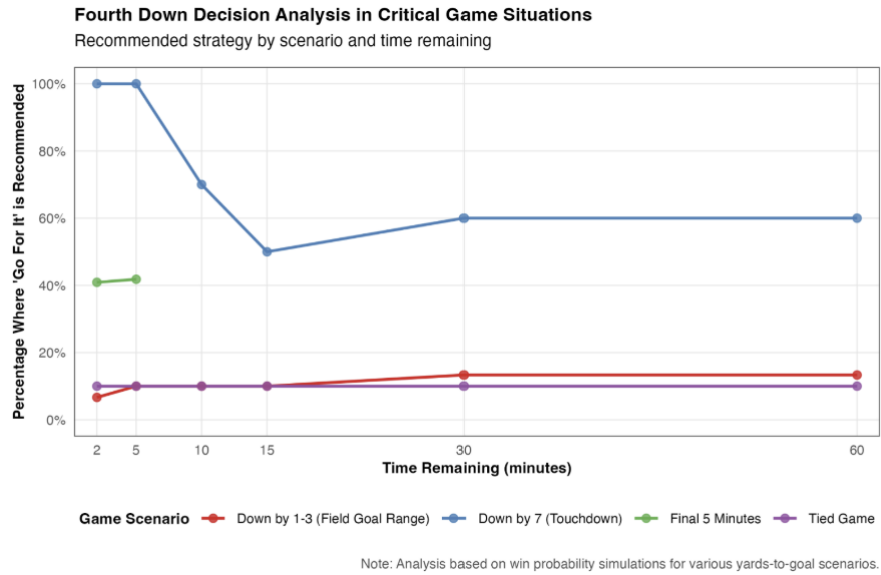


Figure 2: ‘Go for it’ decision analysis with the 5-minute mark (green) showing the critical pivot point in late-game strategy.

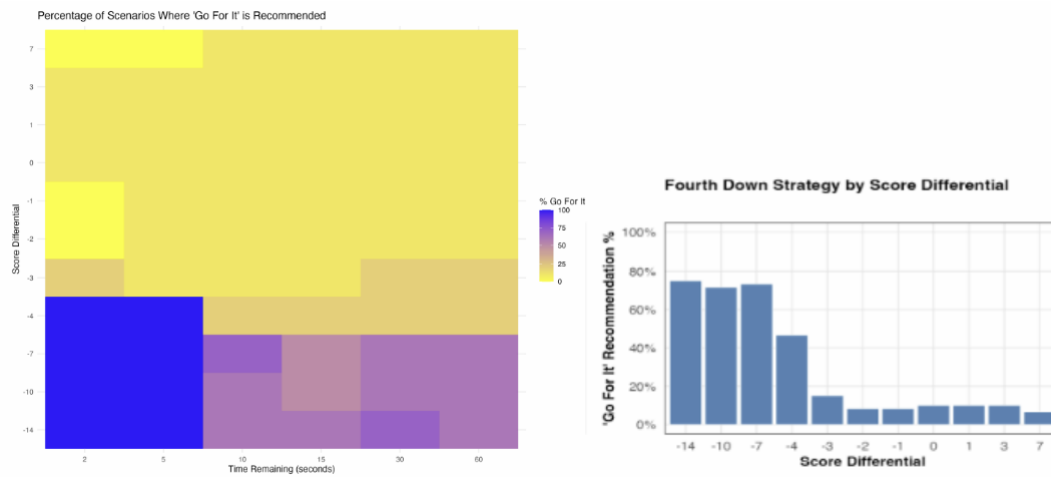


Figure 3: Fourth down strategy analysis showing (1) tactical heatmap by time and score differential and (2) recommendation percentages by score situation.

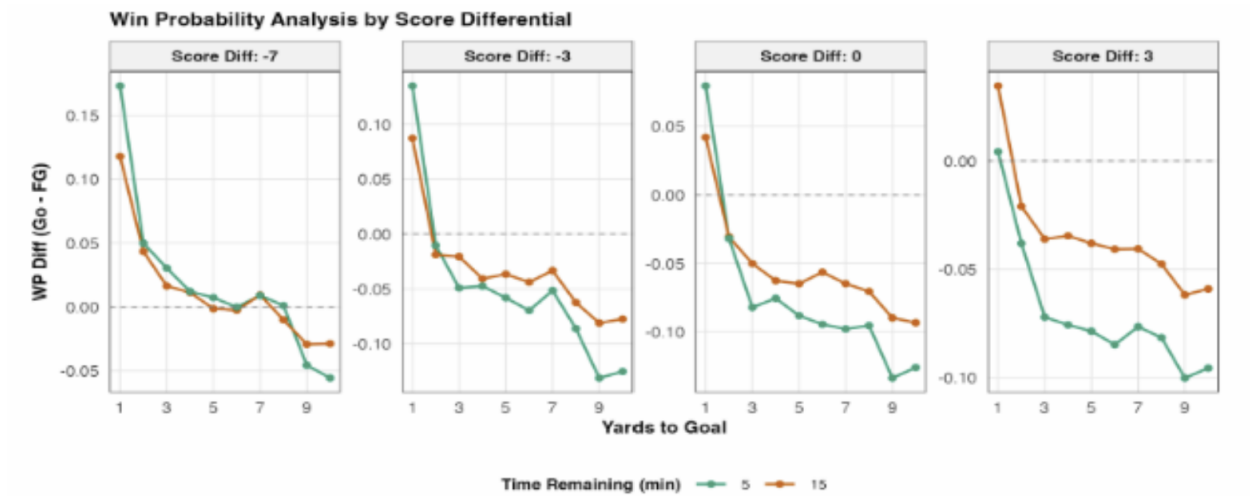


Figure 4: Win probability differential (Go - FG) by yards to goal across score differentials. Green = 5 min remaining; Orange = 15 min remaining.

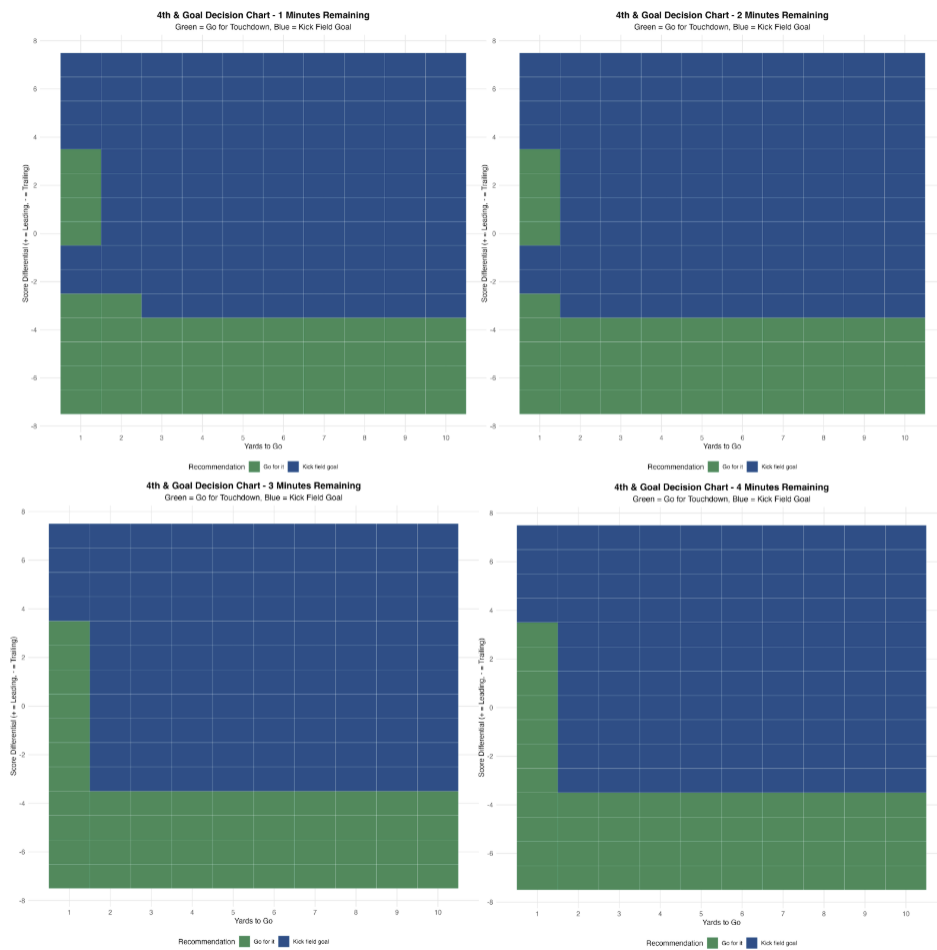


Figure 5: Fourth down decision charts (minutes 1-4). Green = “go for it”; Blue = “field goal”. Score differential (y-axis) vs. yards to goal (x-axis)

4. Discussion

4.1 Simulation Discussion

As discussed previously, this simulation attempts to answer the following question:

In a 4th & Goal situation, in what scenarios should we go for it versus kick a field goal and what factors are necessary for making this decision?

The first interesting part of our simulation was that the score deficit is the primary trigger, especially when the clock is run down to a few minutes left in the game. During any point of the game from 1 minute to 4 minutes left where the score differential was -2.5 or greater, our simulation suggested that we go for it on 4th & Goal. This confirms the real-world principle that you need a touchdown when a field goal still leaves you trailing or tied. Coaches can sense this, but the simulation quantifies this result. However, the majority of decisions suggest that we should be conservative about going for it. In the left panel of Figure 3, we see that the vast majority of the chart is yellow, suggesting that we play it safe most of the time. We enter a gray area when we have 7.5 minutes or greater left on the clock and a score differential of -5.5 or less exists. It seems that most of the time we should go for it, but there is also a smaller probability (~25%) that kicking a field goal and gaining the onside kick may be suggested. This raises questions about which teams should go for onside kicks and which specific scenarios would yield the best outcomes. It also seems that yards-to-go on 4th & Goal matters, but only when it's up to around 3 yards. Your win probability is highest if you "go for it" when the score differential (assuming you're losing on 4th & Goal) is higher and you're ~3 yards away, but as the score differential gets closer to -3, you should go for a field goal. In other words, this gives us a better, concrete cutoff of when your win probability is if you go for it. Throughout the simulation, it seems that the score differential plays the biggest role in 4th & Goal decisions, yardage serves as a secondary lever, the time left on the clock serves as a third lever, and the interaction between these variables brings up interesting edge cases. These insights translate into clearer coaching playbooks and more visibility into statistical decision-making.

Overall, this simulation study challenges the previously held idea that field position (yard to goal) is the most important factor in decision-making. Moreover, it suggests that score differential be considered first, then yards to go, then time remaining. Specifically, when trailing by 4-8 points in the final 5 minutes, teams should almost always go for it regardless of yards to go because a field goal still leaves them trailing.

4.2 Limitations and Future Work

There were certainly limitations while conducting our study, the most notable being the short time constraint as well as a lack of data for 4th & Goal situations. This certainly made us have to decide what we thought we needed to include in our study and what we felt we could omit. If we had more time, we might have incorporated the number of timeouts remaining for each team, as that could certainly have impacted decision-making on these fourth-and-goal situations, especially in end-of-half or late-game scenarios. Also, we could have looked into each NFL team individually, looking at each team's fourth-down success rates to see how teams differ and how that would impact their respective fourth-down decision-making. For example, are there situations where one team should go for it, but another should not? Additionally, our model does not account for the unique offensive and defensive strength of each team; it assumes equal strength for offense and defense for all teams. In reality, this can have a major impact on coaching decisions. For example, a coach who knows they have the best defense in the league that particular season and has a struggling offense may choose to kick a field goal and rely on a defensive stop or interception to get the ball back.

Additionally, our win probability model, while sophisticated, does not account for all game state variables such as weather conditions, home field advantage, or specific matchup dynamics. Also, with more time, we would like to break down plays into pass and run plays and determine how the success rate of each play differs, and how that would, in turn, impact win probability in the fourth-and-goal situation. For example, on 4th and 2 with 5 minutes to go down 4, how do success probabilities differ depending on whether the coach decides to run or pass in this scenario? Additionally, with more time, we could have incorporated the

onside kick success rate and looked into how that may impact fourth-and-goal decision-making. Successfully recovering an onside kick is rather unlikely, but it would be interesting to see if that would have any impact at all on coaching decisions at any point in the game. Without the limitations we faced in this study, incorporating these additional factors would have definitely led to a more well-rounded model when looking at 4th & Goal decision-making on win probability.

Final Thoughts

In conclusion, our simulation provides guidance for coaching decisions in 4th & Goal situations. Overall, there are three main takeaways to remember: 1) Score differential should be the primary consideration, especially in late-game scenarios, with yard to goal secondary; 2) If a team is trailing by 4-8 points late in games (last 5 minutes), they should be considerably more aggressive than if they are trailing by less; 3) The general football principle of “taking the points” when within 3 yards of the goal line early in games is supported by our simulation study, unless there is a significantly high score differential. Overall, these statistically-derived insights can help coaches optimize their decisions, potentially increasing their win probability in close games. Moreover, this simulation study provides the groundwork for further analyses that are highly-specialized to each team’s strengths and weaknesses and analyses that consider other niche, but important aspects of the game (e.g., onside kicks and 2-point conversions).

Appendix A: Code

<https://github.com/andrewtran30/MeanMachine.git>